

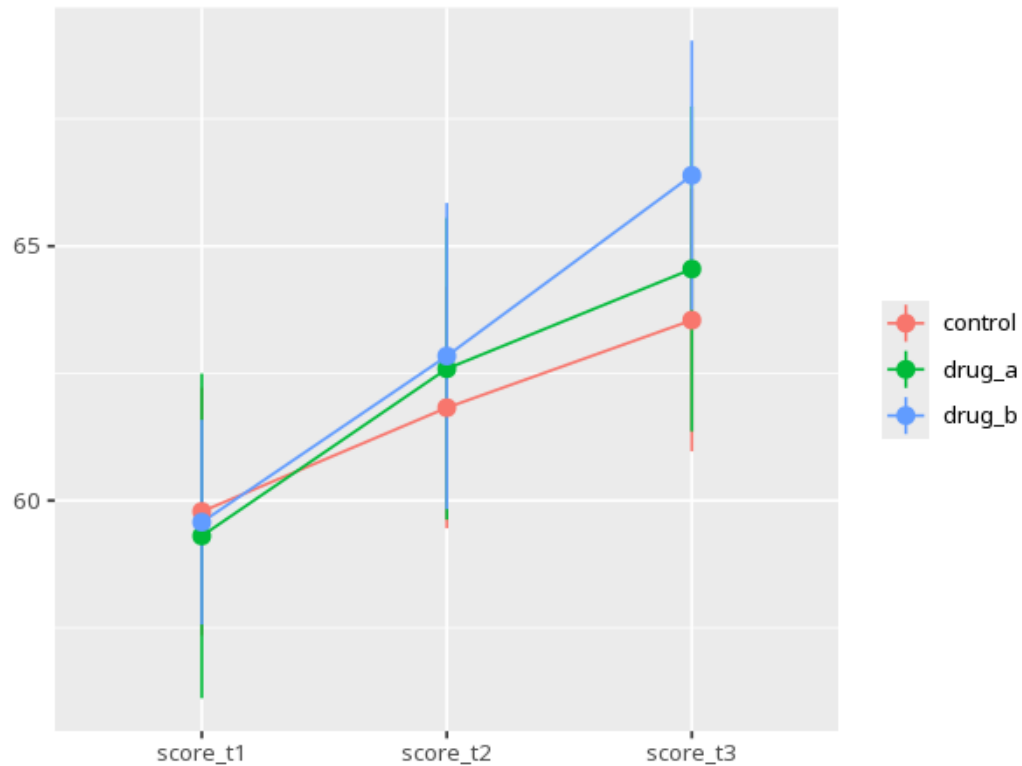
1 Mixed ANOVA

Group	Condition	N	M	SD
control	1	20	59.78	5.21
drug_a	1	20	59.30	6.82
drug_b	1	20	59.58	4.30
control	2	20	61.83	5.06
drug_a	2	20	62.59	6.33
drug_b	2	20	62.84	6.43
control	3	20	63.55	5.51
drug_a	3	20	64.55	6.82
drug_b	3	20	66.39	5.66

Source	SS	df	MS	F	p	partial η^2 [95% CI]
Group (between)	45.68	2	22.84	0.25	.782	0.01 [0.00, 1.00]
Condition (within)	836.30	1.67	500.28	82.56	<.001	0.59 [0.50, 1.00]
Group $\times$ Condition	51.09	3.34	15.28	2.52	.056	0.08 [0.00, 1.00]

Effect	W	p	GG ε	HF ε
condition	0.80	.002	0.84	0.86
grp:condition	0.80	.002	0.84	0.86

Pair	Mdiff	SE	padj	95% CI
score_t1 - score_t2	-2.86	0.42	<.001	[-3.90, -1.83]
score_t1 - score_t3	-5.27	0.32	<.001	[-6.06, -4.49]
score_t2 - score_t3	-2.41	0.48	<.001	[-3.59, -1.23]



Read the Group $\times$ Condition interaction first — a significant interaction means the groups changed differently across conditions; only then read the main effects. Check sphericity for the within and interaction terms — if violated, read the corrected F/p. When the group $\times$ condition interaction is significant, the overall condition comparisons in the post-hoc table can mislead — read each group's line on the profile plot instead. Your significance threshold (α) is 0.05.

A mixed ANOVA yielded a Group $\times$ Condition interaction, $F(3.34, 95.29) = 2.52$, $p = .056$, partial $\eta^2 = .08$ [.00, 1.00].

Statistical basis: Mixed (split-plot) ANOVA (Fisher, R. A. (1925). *Statistical Methods for Research Workers*. Oliver & Boyd.); Mauchly's test of sphericity (rendered above the correction) (Mauchly, J. W. (1940). "Significance test for sphericity of a normal n-variate distribution." *The Annals of Mathematical Statistics*, 11(2), 204-209.); Greenhouse-Geisser correction (sphericity violated) (Greenhouse, S. W., Geisser, S. (1959). "On methods in the analysis of profile data." *Psychometrika*, 24(2), 95-112.); Huynh-Feldt correction (sphericity violated) (Huynh, H., Feldt, L. S. (1976). "Estimation of the Box correction for degrees of freedom from sample data in randomized block and split-plot designs." *Journal of Educational Statistics*, 1(1), 69-82.); Levene's test for equal variances between groups (Brown-Forsythe median-centered variant) (Levene, H. (1960). "Robust tests for equality of variances." In I. Olkin (Ed.), *Contributions to Probability and Statistics* (pp. 278-292). Stanford University Press.); Brown-

Forsythe median-centered variant of the homogeneity-of-variance test (Brown, M. B., Forsythe, A. B. (1974). "Robust tests for the equality of variances." Journal of the American Statistical Association, 69(346), 364-367.); Estimated marginal means for post-hoc comparisons (Lenth RV (2024). emmeans: Estimated Marginal Means, aka Least-Squares Means. R package.).